ML- PROJECT(canteen demand pedictor)

DSAPP Flutter Project

Food Demand & Wastage Prediction System for Canteens

🎯 Project Objective

The goal of this project is to **reduce food wastage in canteens** by tracking daily food preparation and consumption data and using it to **analyze wastage and predict future demand**.

This helps the canteen staff prepare the **right quantity of food**, avoiding both shortages and waste.

🏗️ System Architecture

This project has **three main components**:

1️⃣ Flutter Mobile App (Frontend)

Built using **Flutter (Dart)**.

The app allows users to:

• Add daily food entries
• View history of prepared and consumed food
• Check food wastage summary
• Get prediction of future demand

Screens you implemented:

- **Home Screen**
- **Add Entry Screen**

- **History Screen**
- **Prediction Screen**
- **Summary Screen**

2️⃣ Backend API

Your backend is hosted at:

https://dsappbackend.onrender.com

It provides REST APIs such as:

API	Purpose
`/add-entry`	Add food data
`/history`	Get past records
`/predict`	Predict food demand
`/wastage-summary`	Calculate wastage statistics

3️⃣ Database

The backend stores data in a database containing fields like:

Field	Description
date	Date of entry
item_name	Food item
prepared_qty	Quantity prepared

consumed_qty	Quantity consumed

From this, the system calculates:

wastage = prepared_qty - consumed_qty

⚙️ Features of Your App

1️⃣ Add Food Entry

Users enter:

- Date
- Food item
- Prepared quantity
- Consumed quantity

This data is sent to the backend.

2️⃣ History Tracking

Shows previous records including:

- Food item
- Prepared quantity
- Consumed quantity
- Date

This helps monitor trends.

3️⃣ Wastage Summary

Calculates:

- Average wastage
- Total wastage
- Most wasted food item
- Least wasted item

This helps understand which foods cause waste.

4️⃣ Demand Prediction

Using historical data, the system predicts:

- Which food should be prepared **more**
- Which food should be prepared **less**

This reduces wastage.

📊 Example Workflow

1️⃣ Canteen prepares **100 idlies**
2️⃣ Only **80 are consumed**

Wastage = 20

The system stores this data.

After several days, the prediction model learns patterns and suggests:

Prepare only 85 idlies instead of 100

🛠️ Technologies Used

Frontend

- Flutter
- Dart
- Material UI

Backend

- Python FastAPI

Database

- PostgreSQL

Hosting

- Render Cloud

🌍 Benefits of This System

✔ Reduces food wastage
✔ Saves cost for canteens
✔ Helps in demand planning
✔ Uses data-driven decisions
✔ Easy mobile interface

🚀 Future Improvements

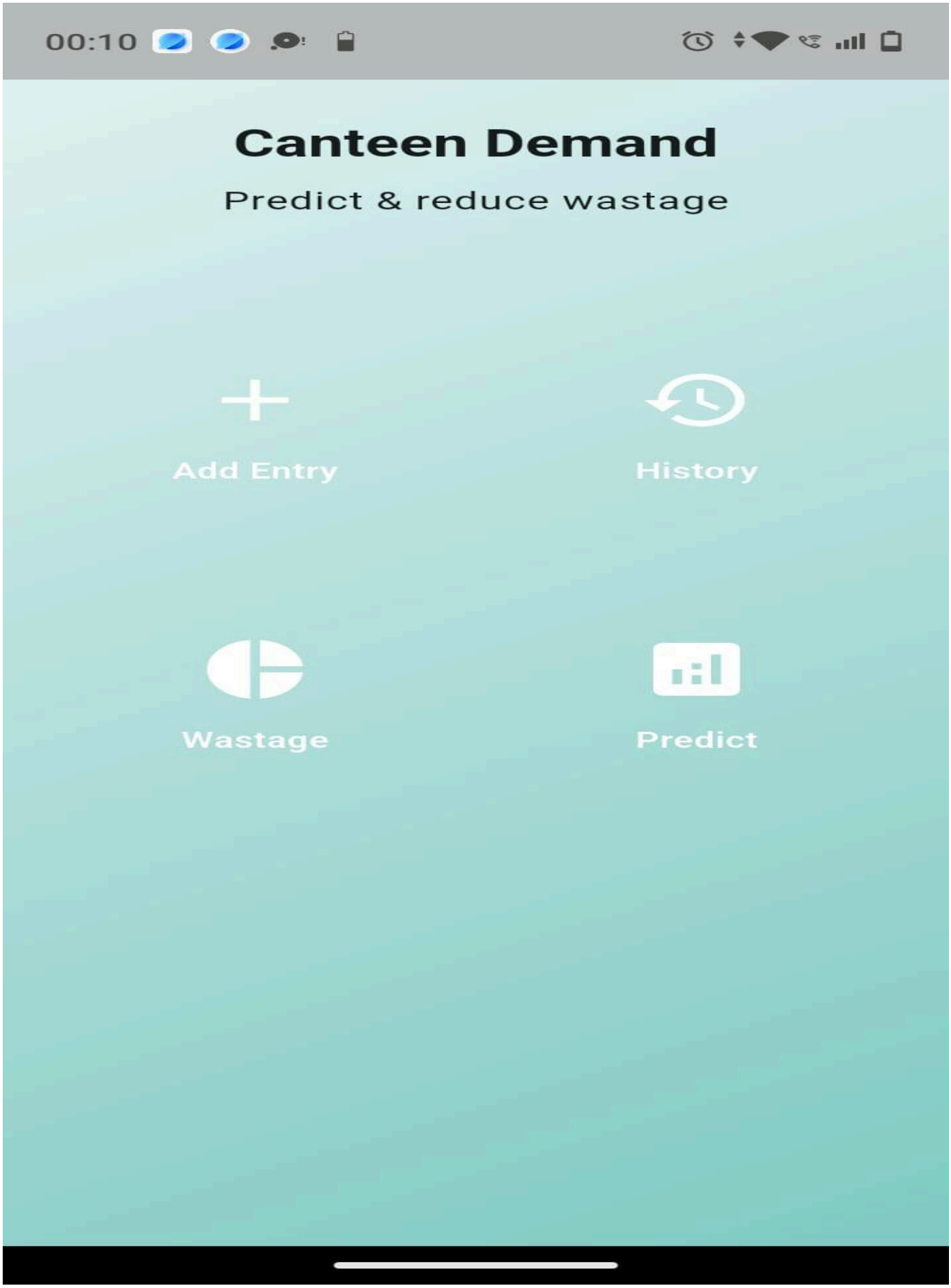
We can add:

• AI/ML demand prediction model
• Graphs for food trends
• Admin dashboard
• Login system
• Notifications for shortages
• Weekly/monthly reports

📌 One-Line Summary (for Viva)

DSAPP Flutter is a mobile application that helps canteens track food preparation and consumption data, analyze wastage, and predict future food demand to minimize food waste.

<u>outputs:</u>


00:10
Canteen Demand
Predict & reduce wastage
Add Entry
History
Wastage
Predict

00:10

Add Entry

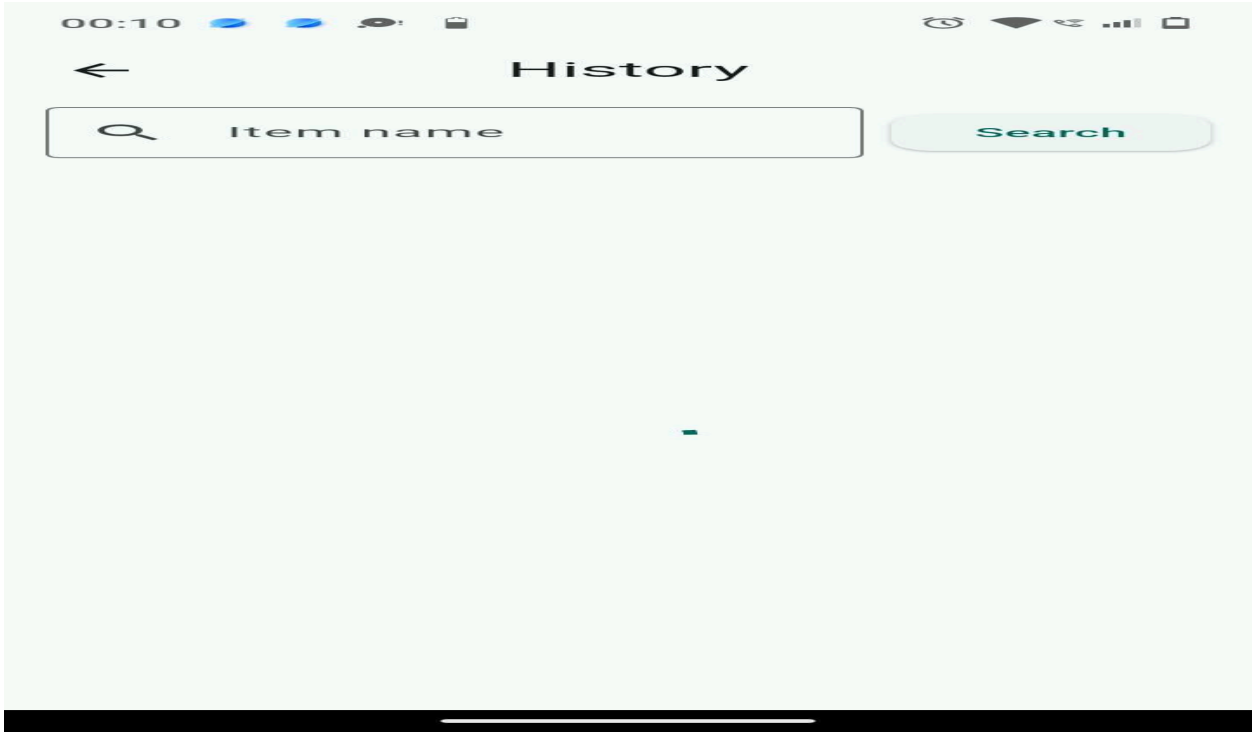
Item name

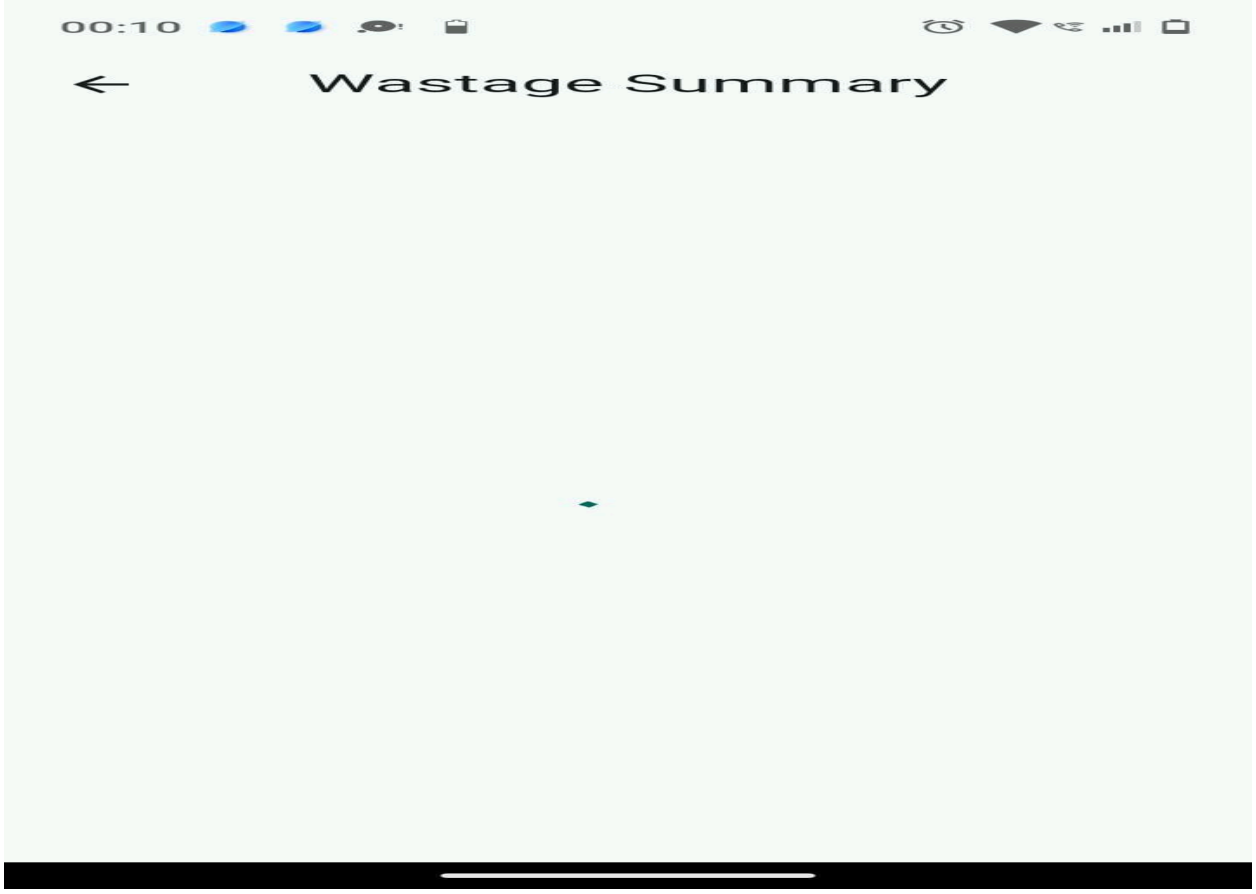
Prepared qty

Consumed qty

Date: 2026-03-08 Pick

Save


00:10
History
Item name
Search


00:10
Wastage Summary

18:26

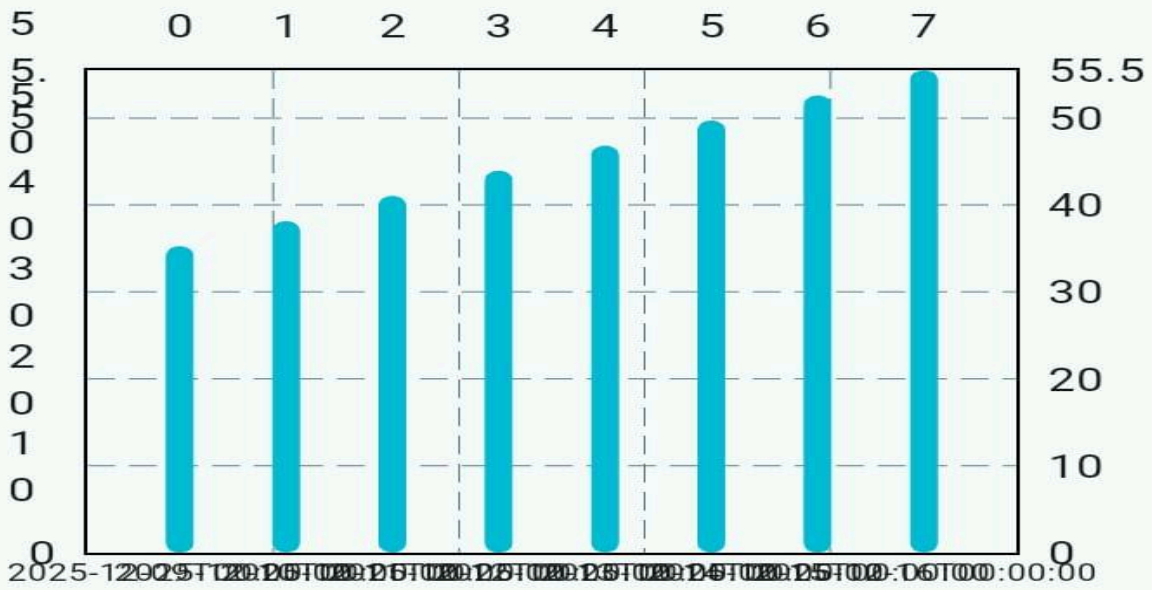
Predict

Item name
vada

Days
8

Go

Chart: Bar



2025-12-09T00:00:00
Predicted consumption
35.27

2025-12-10T00:00:00
Predicted consumption
38.16

2025-12-11T00:00:00
Predicted consumption
41.05

18:25

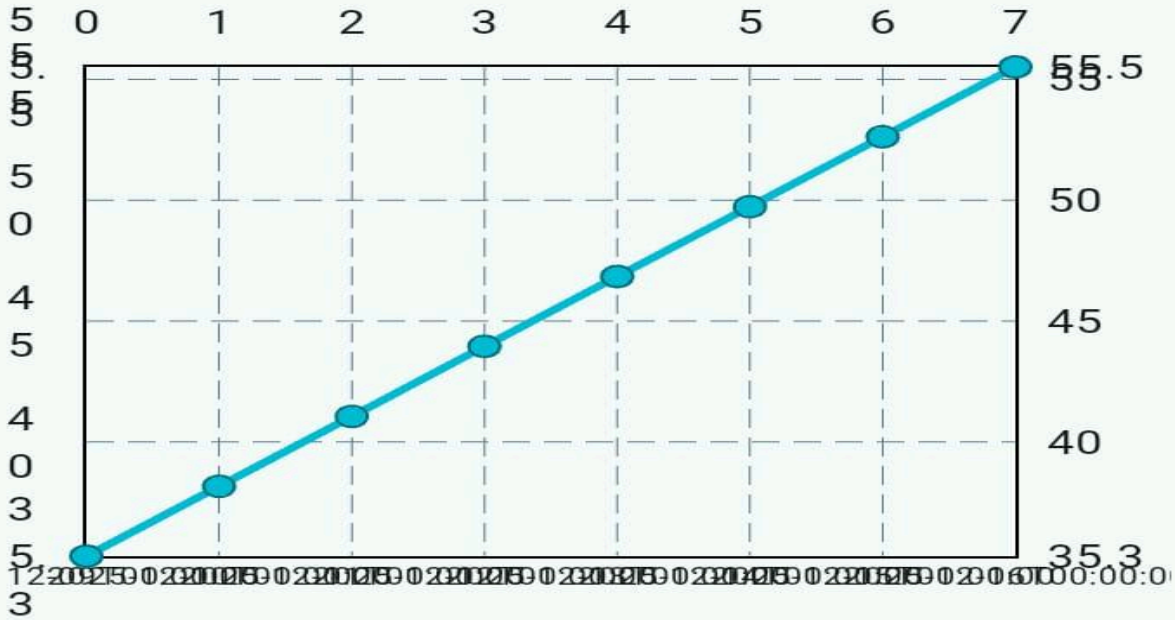
Predict

Item name
vada

Days
8

Go

Chart: Line



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